

Careers in Cyber

TryHackMe Room Report

Introduction to Cyber Security Learning Path

Daaniya Absar

August 9, 2026

Abstract

This report documents my work through the **Careers in Cyber** room on TryHackMe, part of the *Introduction to Cyber Security* learning path. The room provides an overview of the cyber security industry and explores several key roles within the field, including Security Analyst, Security Engineer, Incident Responder, Digital Forensic Examiner, Malware Analyst, and Red Teamer. The purpose of this report is to summarize what each role involves and reflect on which career path best aligns with my own interests and strengths.

Contents

1	Introduction	2
2	Roles Explored	2
2.1	Security Analyst	2
2.2	Security Engineer	2
2.3	Incident Responder	3
2.4	Digital Forensics Examiner	3
2.5	Malware Analyst	3
2.6	Red Teamer	3
3	Careers Quiz	3
4	Reflection	4
4.1	Role That Interests Me Most: Penetration Tester / Red Teamer	4
4.2	Skills I Already Have	4
4.3	Skills I Want to Develop	4
5	Conclusion	5

1 Introduction

The cyber security industry is growing rapidly and offers a wide variety of career paths, ranging from **offensive security** (e.g. penetration testing – legally hacking systems and reporting vulnerabilities) to **defensive security** (e.g. defending against and investigating cyberattacks).

Common reasons to pursue a career in cyber security include:

- **High Pay** – security roles typically offer high starting salaries relative to other IT fields.
- **Exciting Work** – work can include legally hacking systems, hunting threats, and investigating real incidents.
- **High Demand** – organizations across every industry need skilled security professionals, leading to strong job security.
- **Variety of Roles** – there is a role suited to almost every skill set, whether highly technical, analytical, or more strategic and people-focused.

This room breaks down several core roles within the industry, which are explored in the sections below.

2 Roles Explored

2.1 Security Analyst

Security Analyst

Responsible for maintaining the security of an organization's data. Analysts investigate and assess enterprise networks, providing engineers with meaningful data and recommendations to develop preventative security measures. Building strong security measures across an organization is critical to safeguarding it against cyberattacks.

Typical skills: log analysis, threat detection, SIEM tools, incident triage.

2.2 Security Engineer

Security Engineer

Responsible for designing, monitoring, and maintaining security controls, networks, and systems to help prevent cyberattacks. This includes developing security plans, researching new attack tools and trends, and building/hardening secure infrastructure (e.g. implementing firewalls).

Typical skills: networking, system administration, scripting, cloud security.

2.3 Incident Responder

Incident Responder

Responsible for identifying and mitigating attacks while an attacker's operations are still unfolding. The goal is a swift and effective response that retains the organization's financial standing and avoids negative breach implications, ultimately protecting the company's data, reputation, and finances.

Core duties: developing and adopting an actionable incident response plan; maintaining strong security best practices; post-incident reporting and preparing for future attacks based on lessons learned.

2.4 Digital Forensics Examiner

Digital Forensics Examiner

Responsible for collecting digital evidence while observing correct legal procedures, then analyzing that evidence to answer questions related to a case (e.g. how an attacker gained access, what data was affected). This role often works closely with incident responders and legal teams.

Typical skills: forensic imaging, chain-of-custody procedures, evidence analysis, attention to legal/procedural detail.

2.5 Malware Analyst

Malware Analyst

Responsible for analyzing suspicious programs, discovering what they do, and writing reports on their findings. A malware analyst is sometimes called a **reverse engineer**, as their core task involves converting compiled programs from machine language back into readable code, usually in a low-level language such as Assembly.

Typical skills: reverse engineering, disassemblers/debuggers, low-level programming knowledge, sandboxing.

2.6 Red Teamer

Red Teamer

Emulates the role of a threat actor to uncover exploitable vulnerabilities, maintain access, and avoid detection within an organization's systems. Red Teamers assess an organization's security controls, threat intelligence, and incident response procedures, then evaluate and report on their findings with actionable data to help the company prevent real-world attacks.

Typical skills: penetration testing, social engineering, exploit development, stealth/evasion techniques.

3 Careers Quiz

The room includes an interactive quiz designed to help identify which cyber security role best matches an individual's interests and strengths. Since the outcome depends entirely on personal

preferences, there is no fixed "correct" answer – the goal is self-reflection rather than technical accuracy.

Note

After completing the quiz, reflect on the result: does it align with the role you found most interesting while reading through Section 2? Use this as a starting point for choosing which TryHackMe learning path to pursue next.

My Quiz Result

RED TEAMING

4 Reflection

4.1 Role That Interests Me Most: Penetration Tester / Red Teamer

Out of all the roles covered in this room, **Penetration Testing** (closely tied to the Red Teamer role) stands out to me the most. This role involves emulating a real threat actor to actively find and exploit vulnerabilities in an organization's systems, applications, and networks – essentially "legally hacking" a target before a malicious actor can. What appeals to me about this path is the hands-on, problem-solving nature of the work: rather than only monitoring for attacks, a pentester actively probes systems to think like an attacker, uncovering weaknesses before they can be exploited in the real world.

I'm particularly drawn to:

- The offensive, hands-on mindset of finding and chaining vulnerabilities rather than purely defensive monitoring.
- The variety of engagements – web applications, networks, Active Directory environments, and social engineering all fall under this umbrella.
- The clear, tangible outcome of each engagement: a professional report detailing findings and remediation steps that directly improves an organization's security posture.

4.2 Skills I Already Have

- *[e.g. basic understanding of networking concepts (IP addressing, ports, protocols)]*
- *[e.g. comfort using Linux and the command line]*
- *[e.g. foundational knowledge of HTTP and how web applications communicate – from rooms such as HTTP in Detail]*
- *[e.g. strong problem-solving / CTF-style analytical thinking]*

4.3 Skills I Want to Develop

To move toward a Penetration Tester role, the following areas need to be built up next:

- **Networking fundamentals** – deeper understanding of TCP/IP, subnetting, and common network services.
- **Web application security** – the OWASP Top 10 (SQL injection, XSS, IDOR, broken access control, etc.).

- **Linux & scripting** – comfort with Bash/Python for automating recon and exploitation tasks.
- **Enumeration and exploitation methodology** – structured approaches to reconnaissance, vulnerability scanning, and exploitation (e.g. the standard pentest methodology: recon → scanning → exploitation → post-exploitation → reporting).
- **Report writing** – clearly communicating technical findings and their business impact to non-technical stakeholders.

Note

TryHackMe's **Pre Security** and **Jr Penetration Tester** learning paths are the recommended next steps to build these foundations, followed by the **Offensive Pentesting** and **Red Teaming** paths for more advanced, real-world engagement practice.

5 Conclusion

Completing the **Careers in Cyber** room provided a broad overview of the many different career paths available within the cyber security industry, spanning both offensive roles (such as Red Teaming and Penetration Testing) and defensive roles (such as Security Analysis, Incident Response, and Digital Forensics).

Key takeaways from this room include:

- Cyber security is a broad field – there is a role suited to nearly every personality type, whether highly technical, analytical, or strategic.
- Soft skills such as communication, documentation, and collaboration are just as important as technical ability in many roles.
- Starting out does not require knowing everything – picking a path that sparks curiosity and building skills incrementally is the recommended approach.

This room serves as a strong entry point into the *Introduction to Cyber Security* learning path and has helped confirm my own direction: **Penetration Testing**. Going forward, I plan to work through TryHackMe's *Pre Security* and *Jr Penetration Tester* learning paths to build the networking, Linux, and web application security fundamentals needed to pursue this role.

References

- TryHackMe – *Careers in Cyber* room: <https://tryhackme.com/room/careersincyber>
- TryHackMe – *Introduction to Cyber Security* Learning Path